

GENERAL THETA FOUNDATIONS I: CONTROLLED EXPERIMENT DUALITY AND EXACT MARKED MINIMAX LAWS

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ABSTRACT. We study finite-resource statistical comparison when training operations are chosen causally and a validation test is frozen before fresh candidate and target preparations. For finite controlled experiments with a compact parameter space, we prove an architecture-uniform posterior-predictive dual in which action choice remains inside a Bellman recursion and the adversarial prior is chosen only once. The actions need not be garblings of a common experiment. We solve the one-preparation minimax problem for a continuous family of marked product experiments, identifying both an optimal response and a two-point least-favorable prior by an explicit sum-of-squares identity. For the collision target the exact value is $(85 - 7\sqrt{73})/64$. Combined with a deterministic two-preparation audit, this gives an enlarged exact preparation segment at every peak width at least twelve. The microscopic collision and positive-noise bridge are proved within the article. A separate revelation family has the exact joint law $(1 - (1 - \alpha)^N)(1 - 1/K)$ for preparations and decision memory. Compatible finite-architecture certificates and a priced classical autonomous comparison complete the resource analysis.

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1. INTRODUCTION

A statistical experiment determines not only which decisions can be made, but which observations and retained continuations are required to make them. In a controlled experiment, an action may change the information available at the next preparation. In a finite-memory experiment, retaining one continuation may preclude retaining another. A resource theorem must distinguish these two constraints rather than replace both by optimization within one chosen execution graph.

We consider an unknown parameter fixed throughout an audit. Training preparations are conditionally independent draws from causally selected experimental rows. The auditor then freezes a validation design and an event and compares fresh candidate and target draws through that event. All persistent randomness, report buffers and validation accumulators belong to the register. This model separates the simulator's resources from those of its auditor and keeps the mark of the actual preparation inside the statistical comparison.

The first result is a controlled posterior-predictive dual (Theorem 4.1). Its state is an unnormalized posterior measure together with the legal protocol state. At the terminal step, optimization over an event gives a positive-part functional, equivalently a posterior-predictive total variation. Earlier steps optimize the causal experimental action. One minimization over initial priors yields the exact unrestricted-memory minimax value. Every such prior, or a Bellman supersolution, gives a converse uniform over admissible finite-memory architectures. The common-dominating-row theorem becomes a specialization in which the experimental choice can be removed. The controlled formulation also applies when the available actions are statistically incomparable. A strict adaptive advantage persists under full-support perturbations (Proposition 4.4).

Our principal explicit minimax calculation concerns the product family $b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2)$ and targets Q_γ on the four marked diagonal atoms. For $0 \leq \gamma \leq 1 - 1/\sqrt{2}$, put

$$u_\gamma = \gamma - 2 + \sqrt{\gamma^2 - 2\gamma + 5}.$$

Theorem 5.1 gives

$$U_1(b, Q_\gamma) = (3 + u_\gamma^2)/8.$$

The optimal response and least-favorable prior are explicit. Their matching is certified over the entire parameter square by a square identity, not by an optimization grid. The endpoint of the displayed γ interval is the point where a posterior response coefficient changes sign. The zero-preparation value is also exact. At the physical mark bias $\gamma = 1/4$ the one-preparation value is

$$u_{\text{ex}} = (85 - 7\sqrt{73})/64.$$

This resolves the optimization problem hidden behind the earlier rational one-preparation certificate.

The physical consequence uses the same marked collision experiment, not a surrogate alphabet with a different cost convention. Define

$$(1) \quad u_1 = \frac{4035777}{10240000}, \quad m_2 = \frac{12 - 3\sqrt{3}}{16}, \quad \beta_0 = \frac{1}{300}.$$

The older rational bound u_1 remains a valid explicit certificate. The exact value u_{ex} improves it. We prove

$$(2) \quad N_c^{\text{phys}}(W, Q) = 2 \quad \text{for} \quad W \geq 12, \quad \text{TV}(Q, Q_*) \leq \beta, \quad u_{\text{ex}} + \beta \leq c < m_2 - \beta.$$

Here N counts candidate training preparations, and one candidate and one target validation are charged in addition. The upper construction is deterministic and has the complete bit-level profile $(1, 2, 3, 3, 4, 5, 5, 10, 12, 10, 10, 7, 3)$. Definition 6.1 and Theorem 6.3 prove locally every interface and physical approximation fact used in this deduction. Equation (2) is sharp in preparation count on its stated width range; it does not assert width optimality at two preparations or an exact value of U_2 .

A different theorem determines both coordinates in a revelation family. With reveal probability α , N preparations and K retained decision states, Theorem 8.1 proves

$$V_N^{\text{er}}(K) = (1 - (1 - \alpha)^N)(1 - 1/K).$$

The converse allows every stochastic encoder and legal stopping rule; the matching machine requires no uncharged revelation flag. The formula yields exact strict thresholds and matching sample and state scales. The memory coordinate here is a decision-cut width, not the collision machine's bit-level peak. Keeping those coordinates distinct is part of the resource theorem.

For a specified finite architecture, compatible occupation arrays retain all common-row constraints, including autonomous row reuse. The subdivision theorem gives an explicit HS/m certification error with controlled row-block degree. These are effective finite-architecture certificates, not a sharp universal complexity law. We also retain a resource-accounted autonomous binary comparison whose stationary formula is classical. None of these auxiliary results replaces the exact minimax or physical arguments.

1.1. Relation to earlier work. The comparison and minimax mechanisms are classical [2, 3]. Finite-horizon belief-state control and piecewise-linear value functions are established in [1]. Our controlled dual uses these mechanisms with a fixed adversarial parameter, frozen validation and explicit causal resource accounting. It is not a claim to have invented belief-state control or the minimax theorem. The exact marked-family saddle points, their equality certificate and their physical use are stated separately from these general mechanisms.

Polynomial stochastic-policy formulations and positivity certificates have substantial antecedents [4, 5]. Finite-memory testing, randomization and time-invariant rules are treated in [6, 7, 8, 9]. The stationary formula in Section 10 is explicitly attributed to Hellman and Cover. The May 2026 work of Managoli and Prabhakaran [10] treats memory-constrained adversarial binary hypothesis testing, including randomized time-invariant machines and adaptive distribution choices. Its objective and adversarial mechanism differ from the fixed-parameter, training-validation score here. No general priority for randomized state savings is asserted.

Norberg's filtered comparison [11] belongs to the surrounding comparison theory. We make no non-overlap or originality claim about that general theory without a proof-level comparison of the original paper. The novelty claims under consideration are the explicit theorems and resource realizations proved here, not an asserted absence of filtered-experiment predecessors.

1.2. Organization and preserved development. We first fix the architecture and common-row framework, then prove the controlled dual and exact marked minimax law. The physical bridge and charged two-preparation construction give (2). The joint revelation law, finite-architecture certification and autonomous comparison follow. Complete mathematical volumes retain the unaltered predecessor article and development, including the general continuation, Gaussian transport and historical pipeline proofs [12]. These remain mathematical content, not substitutes for a proof in the present article. The independent A2 geometric chain and

the original B4 and broad C2 obligations are not claimed to follow from the finite statistical theorems alone.

2. EXPERIMENTS, ARCHITECTURES AND ATTAINABLE RESOURCES

Throughout, total variation is $\text{TV}(P, Q) = \sup_{0 \leq h \leq 1} (P - Q)h = \frac{1}{2} \sum_x |P(x) - Q(x)|$. An experimental parameter is unknown and fixed throughout an execution. Controller randomization is independent of that parameter. A reset preparation is a fresh draw, independent of earlier preparations given the parameter. A controller may not recover discarded information from an uncharged tape, clock, selector or calibration table.

Definition 2.1 (Finite architecture). A finite architecture A specifies a finite causal execution graph, its legal experimental operations, the observable inputs to each controller row, and a register alphabet at every cut. It also specifies which row occurrences share a stochastic table. A clocked architecture may use different rows at different scheduled times. An autonomous architecture may use time only through a charged state. For a parameter family Θ and terminal reward in $[-1, 1]$, let

$$V(A) = \sup_{q \in \prod_r \Delta_{s_r}} \inf_{\theta \in \Theta} g_\theta(A, q).$$

If $\mathcal{A}(R)$ is the class of architectures obeying a global resource budget R , its outer value is

$$V^{\text{out}}(R) = \sup_{A \in \mathcal{A}(R)} V(A).$$

Both the architecture and the simulator parameter are fixed before the controller's random choices. In particular, a mixture of complete controllers is not automatically an admissible controller at the original register profile.

For a finite hidden execution graph, compatibility can be stated without division. Let $x_{\theta ir}$ be the incoming mass of hidden history i at observable row r , and let $f_{\theta irj}$ be the outgoing mass through choice j . Include the usual fixed-kernel flow equations, as well as

$$(3) \quad f_{\theta irj} \geq 0, \quad \sum_j f_{\theta irj} = x_{\theta ir}, \quad f_{\theta irj} x_{\theta' i' r} = f_{\theta' i' rj} x_{\theta ir}.$$

For shared rows, the indices i, i' include all their occurrences.

Proposition 2.2 (Compatible realization). *For a specified finite architecture, nonnegative flow arrays satisfying (3) are exactly those realized by a common stochastic row at every declared observable input, including at zero incoming mass.*

Proof. Necessity follows from $f_{\theta irj} = x_{\theta ir} q_r(j)$. For sufficiency, if some incoming mass at row r is positive, define $q_r(j)$ by dividing the corresponding outgoing mass by that incoming mass. Conservation makes q_r stochastic, and the cross-products force the same ratio at every other positive incoming mass. If an incoming mass vanishes, nonnegativity and conservation force all its outgoing masses to vanish. If the entire row is unreachable, choose any legal stochastic row. Induction through the finite execution graph reconstructs the prescribed flows. The same ratio argument applies to identified occurrences of an autonomous row. $\square$

This formulation is inherited from the complete companion and is included to fix the controller class. Compact semialgebraic feasibility and an Archimedean positivity theorem provide a complete semialgebraic certificate hierarchy for a *fixed finite architecture*. They do not interchange the two optimizations in the definition. Section 9 gives a quantitative alternative for the inner problem. The next section supplies a different certificate that survives the outer optimization.

3. A STATISTICAL DUAL THAT IS UNIFORM OVER ARCHITECTURES

Let Θ be compact, let Ω be finite, and let $\theta \mapsto P_\theta \in \mathcal{P}(\Omega)$ be continuous. Fix $Q \in \mathcal{P}(\Omega)$. An N -preparation reset selector observes $Z = (Z_1, \dots, Z_N) \sim P_\theta^{\otimes N}$ and chooses a response $h_Z \in [0, 1]^\Omega$. Its expected validation score is

$$(4) \quad G_\theta(h) = \sum_{z \in \Omega^N} P_\theta^{\otimes N}(z)(Q - P_\theta)h_z.$$

The response is fixed before two independent validation draws. Operationally it can be realized by choosing a random event with those coordinate marginals and retaining that event for both validations. No memory bound is imposed in the definition of $U_N(P, Q) = \max_h \min_\theta G_\theta(h)$.

Theorem 3.1 (Posterior-predictive dual). *Both extrema in the following identity are attained:*

$$(5) \quad U_N(P, Q) = \min_{\mu \in \mathcal{P}(\Theta)} \sum_{z, \omega} \left[\int P_\theta^{\otimes N}(z) \{Q(\omega) - P_\theta(\omega)\} d\mu(\theta) \right]_+$$

$$(6) \quad = \min_{\mu \in \mathcal{P}(\Theta)} \sum_z m_\mu(z) \text{TV}(Q, \bar{P}_\mu(\cdot | z)).$$

Here $m_\mu(z) = \int P_\theta^{\otimes N}(z) d\mu(\theta)$ and, when $m_\mu(z) > 0$,

$$\bar{P}_\mu(\omega | z) = \frac{\int P_\theta^{\otimes N}(z) P_\theta(\omega) d\mu(\theta)}{m_\mu(z)}.$$

A zero-mass term in (6) is defined to be zero.

Proof. The response space is a finite-dimensional compact convex box. Probability measures on the compact space Θ form a weakly compact convex set. The integral of (4) is continuous and affine in each argument. The minimum over parameters equals the minimum over probability measures. The minimax theorem [3] therefore interchanges the maximum and minimum. For fixed μ , each response coordinate is independent in the maximization and its contribution is the positive part of its coefficient, proving (5). The coefficient vector at a fixed z has zero sum. For positive $m_\mu(z)$ it equals $m_\mu(z)(Q - \bar{P}_\mu(\cdot | z))$, proving (6). If $m_\mu(z) = 0$, every nonnegative likelihood integrand vanishes μ -almost surely and all coefficients are zero. Continuity on the two compact strategy spaces gives attainment. $\square$

Theorem 3.2 (Dominating-row converse). *Suppose every permitted training row is a parameter-independent causal garbling of one fresh draw from P_θ . Suppose every permitted validation row is the same garbling of the target and candidate laws, and every selected row–event pair is fixed before validation. Then $U_N(P, Q)$ bounds the worst-case expected score of every auditor using at most N candidate training preparations, regardless of its register profile, feedback schedule, independent randomization, or stopping rule. Independent target-only calibration, whose law does not depend on θ , does not invalidate the bound. If the all-on row and all its events are permitted, the bound is attained in the class with no memory cap and N training preparations.*

Proof. Generate N independent all-on observations in advance. At the t th requested preparation, apply the declared garbling to observation t , using only previously simulated reports and independent randomization. This simulates the adaptive transcript with its exact law. A stopped execution ignores the unused observations; it does not reveal them to the auditor. Pull back its final row–event response to the all-on alphabet. Conditional on the entire generated training word and the independent random seed, this is a function in $[0, 1]^\Omega$. Averaging the seed, including any target-only calibration, produces one response h_z for each full word z . The score is consequently (4). This proves the upper bound for the union of all architectures. Conversely,

retaining the entire word and then a selected event implements any such finite response kernel when memory is unrestricted. Fresh validation gives exactly (4). $\square$

Remark 3.3 (What the dual does not convexify). The measure μ is an adversarial prior used to bound the minimum over *fixed* parameters. Its likelihood is $\int P_\theta^{\otimes N} d\mu(\theta)$, not $(\int P_\theta d\mu(\theta))^{\otimes N}$. It neither enlarges the private candidate class to mixtures nor gives the auditor a free persistent selector. Attainment of U_N without a memory cap does not assert attainment at any specified finite profile.

The state in (6) is generated by the actual preparation law, not by a geometric measure chosen independently of the experiment. For sequential continuation one retains the posterior μ_z or its future-test equivalence class; a single predictive marginal need not be closed under further observations. This is the distinction between a one-step response quotient and a future-predictive quotient.

Corollary 3.4 (Finite moment support for the product family). *For two independent Bernoulli report bits with parameters (p, q) and an independent fair mark, an optimal prior in (5) can be chosen with at most $(N + 2)^2$ atoms. The two report counts (k, ℓ) are a causally updatable sufficient statistic for computing the dual.*

Proof. Conditional on the two counts, all arrangements of report bits and all mark words have parameter-independent probabilities. Summing their coefficients reduces (5) to the $(N + 1)^2$ count pairs. Their likelihoods are

$$\binom{N}{k} p^k (1 - p)^{N-k} \binom{N}{\ell} q^\ell (1 - q)^{N-\ell}.$$

After multiplication by $Q(\omega) - P_{p,q}(\omega)$, every integrand has degree at most $N + 1$ in each variable. Thus the objective depends only on the $(N + 2)^2$ rectangular moments. Their constant coordinate is one. Carathéodory's theorem in the remaining affine space replaces an optimal moment vector by a convex combination of at most $(N + 2)^2$ point evaluations. Counts update by addition of the current bits. No claim that this count register is optimal for the decision is made. $\square$

4. CONTROLLED EXPERIMENTS WITHOUT A DOMINATING ACTION

The reduction to one dominating row is not needed for a statistical dual, provided causal action choice is retained in the dual recursion. We give the finite-action theorem with a compact parameter space and an explicit protocol state. Its equality is an unrestricted-memory statement; its upper certificates apply to every smaller-memory class.

Let Θ be compact metric and E a finite set of protocol states. At $e \in E$ let $\mathcal{A}_e$ be a finite set of training actions. Action a produces x in a finite alphabet X_a with law P_θ^a and changes the protocol state to $T(e, a, x)$. The functions $\theta \mapsto P_\theta^a(x)$ are continuous. An unknown θ is fixed throughout the experiment. Conditional on it, each requested preparation uses a fresh independent draw from the selected row. Legal validation designs form a nonempty finite set $\mathcal{B}_e$. Design b has a finite outcome alphabet Y_b , target law Q^b and continuous candidate law V_θ^b . A policy may stop after at most N training actions. It then freezes $b \in \mathcal{B}_e$ and an event on Y_b , before one candidate and one target validation. Randomization is independent of θ . No training action is assumed to dominate another.

For a finite nonnegative measure ν on Θ write $\nu_{a,x}(d\theta) = P_\theta^a(x)\nu(d\theta)$ and set

$$(7) \quad \Phi_e(\nu) = \max_{b \in \mathcal{B}_e} \sum_{y \in Y_b} \left[Q^b(y)\nu(\Theta) - \int V_\theta^b(y) d\nu \right]_+,$$

$$F_0(e, \nu) = \Phi_e(\nu),$$

$$(8) \quad F_n(e, \nu) = \max \left\{ \Phi_e(\nu), \max_{a \in \mathcal{A}_e} \sum_{x \in X_a} F_{n-1}(T(e, a, x), \nu_{a,x}) \right\}.$$

If no training action is legal, only the stopping term is retained. All the functions vanish at the zero measure, so no posterior division is required on null histories. For positive mass, (7) is the mass times the largest posterior-predictive total variation.

Theorem 4.1 (Controlled posterior-predictive dual). *Let $\mathcal{U}_N(e_0)$ be the maximum worst-case expected validation score over all randomized causal policies in the declared protocol, with at most N training preparations and no memory cap. Then*

$$(9) \quad \mathcal{U}_N(e_0) = \min_{\mu \in \mathcal{P}(\Theta)} F_N(e_0, \mu).$$

Both extrema are attained. Each $F_n(e, \cdot)$ is the maximum of finitely many continuous linear functionals on the cone of measures; in particular it is convex, continuous and positively homogeneous. For every prior μ , $F_N(e_0, \mu)$ bounds the worst-case score of every finite-memory architecture implementing this protocol.

Proof. There are finitely many histories of length at most N and finitely many event masks at each stopping history. Consequently there are finitely many deterministic policy trees, denoted by $\Pi_N(e_0)$. For $\pi \in \Pi_N(e_0)$ its payoff $g_\pi(\theta)$ is a finite sum of finite products of continuous primitive probabilities and is continuous in θ .

A randomized causal policy induces the same outcome laws as a probability distribution on these trees: sample its finite random decision table in advance and condition on it. Conversely, with no memory cap, a sampled tree can be retained and followed. Thus the unrestricted payoff vectors are exactly the convex hull of the g_π . This step is not an assertion that mixing machines preserves a specified register width.

The simplex on $\Pi_N(e_0)$ and $\mathcal{P}(\Theta)$ are compact and convex. The expected payoff is continuous and bilinear. Minimax therefore gives

$$\max_{\lambda \in \Delta(\Pi_N)} \min_{\theta} \sum_{\pi} \lambda_{\pi} g_{\pi}(\theta) = \min_{\mu} \max_{\pi \in \Pi_N} \int g_{\pi} d\mu.$$

For a fixed measure, the terminal maximum is (7), because an event includes precisely the positive coordinates of its signed predictive measure. At a training node one selects an action and, after its observed outcome, selects a continuation tree separately on that branch. The unnormalized branch measure is exactly $\nu_{a,x}$. Backward induction proves (8) and identifies the right side with (9). It also proves the finite-linear-functional representation. Compactness gives an optimizing prior and an optimizing mixed tree. Finally, the minimum over fixed parameters of any policy is at most its average under μ , which is at most $F_N(e_0, \mu)$. This remains true after restricting its memory or architecture. $\square$

The minimization in (9) occurs once, outside the recursion. It cannot be replaced by fresh adversarial choices after observations. The posterior carries dependence on one fixed unknown parameter. Also, a Bayes-optimal tree for a minimizing prior need not by itself be a minimax policy; the saddle-point mixture is supplied by the finite game in the proof.

Corollary 4.2 (Bellman upper certificates). *Suppose functions $\Psi_n(e, \nu)$ satisfy $\Psi_0 \geq \Phi$ and, for $n \geq 1$, satisfy*

$$\Psi_n(e, \nu) \geq \Phi_e(\nu), \quad \Psi_n(e, \nu) \geq \sum_x \Psi_{n-1}(T(e, a, x), \nu_{a,x}) \quad (a \in \mathcal{A}_e).$$

Then $\Psi_N(e_0, \mu)$ is an upper certificate for every admissible architecture and every prior μ . The family F_n is the least such family. An optimal prior can be chosen with at most $d + 1$ support points, where d is the affine dimension of $\{(g_\pi(\theta))_{\pi \in \Pi_N} : \theta \in \Theta\}$.

Proof. Induction in (8) gives $F_n \leq \Psi_n$. The equality case is the recursion itself. The objective depends on a prior only through its vector of integrals of the finitely many g_π . An optimal such vector belongs to the convex hull of their compact evaluation image. Carathéodory's theorem gives the stated support bound while preserving every pure-tree payoff. $\square$

Proposition 4.3 (Causal stability). *Consider two models with the same protocol, alphabets and policy class. Suppose their training rows differ uniformly in total variation by at most ϵ , their candidate validation rows by at most η , and their target validation rows by at most β . Their unrestricted values, or their values in any common specified memory class, differ by at most $2N\epsilon + \eta + \beta$.*

Proof. Use the same policy randomization and couple corresponding training rows while the histories agree. The probability of a first discrepancy is at most $N\epsilon$, even with feedback and stopping. Conditional expected validation rewards lie in $[-1, 1]$, so differing training histories contribute at most $2N\epsilon$. On agreeing histories the same frozen design and event are used, and their conditional mean rewards differ by at most $\eta + \beta$. The bound is uniform in the fixed parameter and in the policy. Taking minima and suprema preserves it, with the reverse inequality obtained by interchanging the models. $\square$

4.1. An informative action choice. The following calculation distinguishes the controlled theorem from a common-row reduction. Let $\Theta = \{(i, j) : i, j \in \{0, 1\}\}$. Action R reports i . Action L reports j when $i = 0$ and otherwise reports an independent fair bit. Action H reports j when $i = 1$ and otherwise reports an independent fair bit. The candidate validation is $\delta_{(i,j)}$ and the target is uniform on Θ .

Proposition 4.4 (Strict adaptive advantage). *With two training preparations, the unrestricted adaptive value is $3/4$ and the best nonadaptive value is $5/8$. Nonadaptive policies here include mixtures of schedules, with their random schedule retained. No available single training action dominates all the others.*

Proof. Take R first, and then L or H according to the observed i . The parameter is recovered exactly. The complement of its validation atom has score $3/4$. No policy can exceed $\text{TV}(Q, \delta_\theta) = 3/4$ even with full information.

For the converse for fixed schedules, use the uniform prior. For a pair of actions a, b its Bayes value is explicitly

$$(10) \quad \sum_{x,y,\theta} \left[\frac{1}{16} \sum_{\vartheta} P_\vartheta^a(x) P_\vartheta^b(y) - \frac{1}{4} P_\theta^a(x) P_\theta^b(y) \right]_+.$$

Substitution of the three displayed kernels gives

	R	L	H
R	$1/2$	$5/8$	$5/8$
L	$5/8$	$7/16$	$1/2$
H	$5/8$	$1/2$	$7/16$

This is an exact finite sum, not a parameter-grid estimate. It bounds any mixture of fixed schedules by $5/8$; an early stop can be padded with an ignored observation. To attain the bound uniformly in θ , choose with equal probabilities the schedules (R, L) and (R, H) . When the chosen branch-specific action matches i , the parameter is known and the score is $3/4$. Otherwise only i is known, and the complement of its two-atom fiber gives score $1/2$. Each fixed parameter therefore has mean score $5/8$.

Action R cannot distinguish parameters differing only in j . Action L cannot distinguish $(1, 0)$ and $(1, 1)$, and H cannot distinguish $(0, 0)$ and $(0, 1)$. An experiment which fails to distinguish such a pair cannot garble to one that distinguishes it. Hence no action is a common dominating row. $\square$

If every binary report is additionally passed through an independent binary symmetric channel of crossover ϵ , the primitive training rows change by at most ϵ and have full support for $0 < \epsilon < 1/2$. Proposition 4.3, applied to both policy classes, leaves an adaptive advantage at least $1/8 - 8\epsilon$, which is positive for $0 < \epsilon < 1/64$. Thus the distinction is not confined to disjoint observation supports. The finite-horizon belief recursion and its piecewise-linear structure are classical [1]. The contribution here is their explicit integration with the frozen validation score, one fixed adversarial parameter, outer certificates, and the resource conventions; no priority for Bellman recursion itself is claimed.

5. EXACT MINIMAX LAWS FOR A MARKED PRODUCT FAMILY

The posterior-predictive dual can be solved explicitly for a continuous family containing the collision target. Throughout this section, the training and candidate validation law is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad (p, q) \in [0, 1]^2.$$

In the order 000, 001, 110, 111, let Q_γ have masses

$$((1 + \gamma)/4, (1 - \gamma)/4, (1 - \gamma)/4, (1 + \gamma)/4),$$

and zero mass elsewhere. Write $U_1(\gamma) = U_1(b, Q_\gamma)$, with exactly the training-validation convention of (4). Set

$$(11) \quad \gamma_* = 1 - 1/\sqrt{2}, \quad s_\gamma = \sqrt{\gamma^2 - 2\gamma + 5}, \quad u_\gamma = \gamma - 2 + s_\gamma, \quad \tau_\gamma = \frac{2u_\gamma}{u_\gamma + 2 - \gamma}.$$

Theorem 5.1 (Exact marked minimax family). *For every $0 \leq \gamma \leq \gamma_*$,*

$$(12) \quad U_1(\gamma) = \frac{3 + u_\gamma^2}{8}.$$

An optimal response depends on the training word only through the report count $K = X_1 + X_2$. On the four target atoms its rows are

$$(13) \quad h_0 = (\tau_\gamma, 0, 1, 1), \quad h_1 = (1, 1, 1, 1), \quad h_2 = (1, 1, 0, \tau_\gamma),$$

and all responses are zero off these atoms. A least-favorable prior is

$$(14) \quad \mu_\gamma = \frac{1}{2}\delta_{(t_\gamma, t_\gamma)} + \frac{1}{2}\delta_{(1-t_\gamma, 1-t_\gamma)}, \quad t_\gamma = \frac{1 - \sqrt{u_\gamma}}{2}.$$

The assertion optimizes over all stochastic responses to the full marked training word, not merely over count-based or symmetric responses.

Proof. First, u_γ is the positive root of

$$(15) \quad u^2 + (4 - 2\gamma)u - (1 + 2\gamma) = 0.$$

On the asserted interval, $0 < u_\gamma < 1$ and $0 < \tau_\gamma < 1$. Indeed the polynomial is negative at zero and positive at one; also $u_\gamma < 2 - \gamma$ because $\gamma < 1/2$. Substitution of $1 - 2\gamma$ into the polynomial gives $8(\gamma^2 - 2\gamma + 1/2)$, which is nonnegative precisely on the present interval. Monotonicity on $[0, \infty)$ therefore gives

$$(16) \quad u_\gamma \leq 1 - 2\gamma.$$

Put $a = (1-p)(1-q)$, $c = pq$, and $z = 1 - a - c$. The probabilities of the three training counts are a, z, c . The independent training mark may be discarded by the displayed construction. Let $G_\gamma(p, q)$ denote its score. With $u = (p + q - 1)^2$ and $v = (p - q)^2$, direct expansion gives

$$(17) \quad G_\gamma(p, q) - \frac{3 + u_\gamma^2}{8} = \frac{2 - \tau_\gamma}{16} \{(u - v - u_\gamma)^2 + 4v\}.$$

For clarity, this identity follows by substituting the three rows of (13) into

$$a(Q_\gamma - b)h_0 + z(Q_\gamma - b)h_1 + c(Q_\gamma - b)h_2,$$

using $a + c = (1 + u - v)/2$, $a - c = -(p + q - 1)$, and then eliminating u_γ^2 by (15). The coefficient of the resulting square is $(2 - \tau_\gamma)/16$ and that of v is four times this coefficient. Both are nonnegative. Equality holds at both points of (14), since there $v = 0$ and $u = u_\gamma$. This proves the lower bound and identifies its equality set.

It remains to bound every response, including nonsymmetric ones. Use (14) in Theorem 3.1. At $t = t_\gamma$ put $a = (1 - t)^2$, $c = t^2$, and $T = a + c = (1 + u_\gamma)/2$. For the training report pair 00, the four dual coefficients, apart from a common positive factor, are

$$(18) \quad \left(Q_H T - \frac{a^2 + c^2}{2}, \quad Q_L T - \frac{a^2 + c^2}{2}, \quad Q_L T - ac, \quad Q_H T - ac \right),$$

where $Q_H = (1 + \gamma)/4$ and $Q_L = (1 - \gamma)/4$. The first coordinate is zero by (15); the second is $-\gamma T/2 \leq 0$. The last two are positive: $T \geq 1/2$, $ac \leq 1/16$, and $\gamma < 1/2$ imply $Q_L T > 1/16 \geq ac$. Thus $(\tau_\gamma, 0, 1, 1)$ maximizes this linear functional. For the training pair 11, reversing all report bits and the mark reverses the four coordinates and gives the row h_2 .

For either training pair 01 or 10, the two prior support points have the same likelihood $t(1 - t)$. The posterior is still equally weighted, and the candidate predictive mass at each target atom is $T/4$. Every one of the four coefficients is therefore nonnegative by $Q_L - T/4 = (1 - 2\gamma - u_\gamma)/8 \geq 0$. The response h_1 is optimal. At off-diagonal validation atoms the target mass vanishes and the coefficient is nonpositive. Finally, conditioning on either training mark multiplies all these coefficients by $1/2$; it creates no further optimizing choice. We have therefore optimized every coordinate of the full marked-word response space.

The prior average of the displayed response equals its value at either support point, namely $(3 + u_\gamma^2)/8$, by (17). Every other response has prior average at most this number and hence has minimum over fixed (p, q) at most this number. Together with the lower bound this proves (12), optimality of the response, and least favorability of the prior. $\square$

Corollary 5.2 (The exact one-preparation collision value). *For $Q_* = Q_{1/4}$,*

$$(19) \quad U_1(b, Q_*) = u_{\text{ex}} := \frac{85 - 7\sqrt{73}}{64}, \quad u_{1/4} = \frac{\sqrt{73} - 7}{4}, \quad \tau_{1/4} = 2 - \frac{14}{\sqrt{73}}.$$

If only the target changes to Q , then $|U_1(b, Q) - u_{\text{ex}}| \leq \text{TV}(Q, Q_)$.*

Proof. Substitute $\gamma = 1/4$ in Theorem 5.1. For a fixed trained response the target perturbation is at most total variation, uniformly in (p, q) . Taking first the minimum and then the maximum preserves this bound in both directions. $\square$

5.1. Complementary slackness and the event alphabet. The tie in the first coordinate of (18) explains the randomization. Any value of that response coordinate is Bayes-optimal for the two-point prior. The minimax choice makes the uniform payoff stationary at its two support points; solving this condition gives τ_γ and the square identity (17). This derives the prior and response together, rather than postulating that a numerically effective prior is optimal.

Operationally, for $K = 0$ one draws once between the events with indicator vectors $(0, 0, 1, 1)$ and $(1, 0, 1, 1)$, with probabilities $1 - \tau_\gamma$ and τ_γ . For $K = 2$ one draws between $(1, 1, 0, 0)$ and $(1, 1, 0, 1)$. For $K = 1$ the event is $(1, 1, 1, 1)$. The resulting event identity is retained and used for both fresh validations. These are exactly the five events in the two-preparation construction below. The one-preparation theorem supplies their complementary-slackness interpretation; it does not prove that the two-preparation selector is optimal. The four-point rational prior retained below is a useful rational certificate, not a least-favorable prior.

The exact response uses an atomic stochastic row. It does not receive free persistent randomness. Replacing τ_γ by a dyadic number within 2^{-b} changes each conditional response, and hence the score, by at most 2^{-b} . A sequential fair-bit implementation must retain its bit prefix and event identity. None of these implementation costs is needed by the deterministic two-preparation physical audit.

Proposition 5.3 (The zero-preparation endpoint). *For $0 \leq \gamma \leq 1$, $U_0(b, Q_\gamma) = \gamma/2$.*

Proof. The fixed event $\{000, 111\}$ has score $(1 + \gamma)/2 - (a + c)/2 \geq \gamma/2$ because $a + c \leq 1$. Conversely, the equally weighted prior on $b(0, 0)$ and $b(1, 1)$ has predictive validation law uniform on the four target atoms. Its total variation from Q_γ is $\gamma/2$. Theorem 3.1 with $N = 0$ gives the matching upper bound. $\square$

6. A SELF-CONTAINED COLLISION INTERFACE

We give the preparation, mechanical estimate, observation law and private simulator class needed for the physical specialization. The following argument does not appeal to an unpublished physical lemma. It retains the same positive-volume preparation and positive sensor noise as the earlier collision construction.

Definition 6.1 (The reset row–event architecture). A preparation supplies two sequential report slots and, afterwards, an actual mark. The source reports in the two slots are constant. A private width-one simulator has no retained information between those slots and no persistent hidden seed. It may use fresh independent coins. Its first emission is $X_1 \sim \text{Ber}(p)$; after receiving the current gate $A \in \{0, 1\}$, its second emission has a fixed Bernoulli kernel with parameter q_A . The parameters (p, q_0, q_1) are fixed throughout the audit. The simulator does not observe the actual mark. All its coins are independent of that mark. Different preparations reset all unretained variables and are independent conditional on the parameters.

The auditor observes the first report, chooses a gate using its retained state and that report, observes the second report, and then observes the mark. During training it may choose rows and stop adaptively, using at most the declared number of candidate preparations. It next freezes one gate policy and one event, with any retained random choice included in its register. It uses that same row–event continuation on one fresh candidate preparation and one independent target preparation and returns the difference of their event indicators, target minus candidate. It may not reselect the event after either validation. Independent target-only calibration is allowed but all retained calibration and randomness are charged. Peak width counts every bit, mark, event and validation-accumulator cut. Clocked schedules are declared externally; an autonomous implementation must retain its phase. An unretained transcript, seed or stopping-time value is not available to a continuation.

In the all-on row the complete private law is $b(p, q_1)$, with an independent fair actual mark. For lower bounds it suffices to restrict the adversary to $q_0 = 0$, $q_1 = q$. In this subfamily every deterministic gate policy is the pushforward

$$T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w)$$

of $b(p, q)$. The gate, when included in the transcript, is another function of these same coordinates. Randomized gate policies are covered by averaging their independent random tables. This proves both the all-on product characterization for the upper construction and the common-garbbling subfamily used for the outer converse. The width-one constraint is a simulator constraint; it is not an uncharged restriction on the auditor's register.

6.1. Preparation and elastic scattering. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$, with no walls and contact distance d . Write $r = q_1 - q_2$, $g = v_1 - v_2$, $V = (v_1 + v_2)/2$, and let the center of mass position range over any fixed rational box of positive volume. Prepare $(B, W) \in \{0, 1\}^2$ with probability $5/16$ at $B = W$ and $3/16$ at $B \neq W$. Conditional on (B, W) , use the uniform product preparation on the following boxes:

$$(20) \quad \begin{aligned} r &= (-a_0, (2B-1)b_0, c_0), & a_0 &\in [29/10, 31/10], & b_0 &\in [9/20, 11/20], \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & & & |V_z - (2W-1)| &\leq 1/100. \end{aligned}$$

Thus the actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$, preserved by momentum conservation. Set

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector, identically zero on the incoming transverse-velocity range.

Lemma 6.2 (Uniform collision signal). *For every preparation in (20) and every $9/10 \leq d \leq 11/10$, exactly one collision occurs and its time lies in $(4/5, 3/2)$. Afterwards*

$$(2B-1)v_{1,y} > 1/2, \quad (2B-1)h(v_{1,y}) > 1/4.$$

For $0 < d \leq 2/5$ no collision occurs on $[0, 3]$ and the detector is zero throughout that interval.

Proof. Before contact, $r(t) = r + tg$. At $t \leq 4/5$ its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At $t = 3/2$ its absolute coordinate bounds are $1/4$, $113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. The first contact therefore occurs in the claimed interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$. Hence the entry contact has $n_x < 0$, where $n = r(t_c)/d$. Its remaining coordinates give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B-1)n_y \geq 87/220.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The elastic rule gives

$$v_1^+ = V + g/2 - (g \cdot n)n, \quad g^+ = g - 2(g \cdot n)n.$$

Consequently

$$(2B-1)v_{1,y}^+ > -3/200 + (1443/1000)(87/220) > 1/2.$$

The definition of h implies the signal bound. Since $r(t_c) \cdot g^+ > 0$, squared separation increases strictly afterwards; there is no second collision. Finally, for $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. There is no contact, and the incoming $|v_{1,y}| \leq 3/200$ makes h zero. $\square$

6.2. Positive noise and the actual mark. Let $\mathcal{B}_t$ be Brownian motion independent of the preparation and set

$$Y_t = \int_0^t h(v_{1,y}(s)) ds + \sigma \mathcal{B}_t, \quad 0 < \sigma \leq 1/24.$$

The two potential all-on reports are

$$Z_1 = \mathbf{1}_{\{2(Y_2 - Y_{3/2}) > 0\}}, \quad Z_2 = \mathbf{1}_{\{2(Y_3 - Y_{5/2}) > 0\}}.$$

The actual second report is AZ_2 . The gate changes acquisition, not the mechanical path. The continuous path is implemented by the target sensor; it is not made available to the private simulator or supplied as an uncharged auditor buffer. Denote the law of (Z_1, Z_2, W) by $Q_{d,\sigma}$.

Theorem 6.3 (Uniform physical presentation). *For $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$,*

$$(21) \quad \text{TV}(Q_{d,\sigma}, Q_*) < 1/300.$$

The same estimate holds after every declared feedback gate, including its observable gate value. The all-on private product family and the common-garbling subfamily in Definition 6.1 apply to this very experiment.

Proof. Every collision occurs before $3/2$. Conditional on the preparation, the two normalized sensor windows are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. Lemma 6.2 makes their means have sign $2B - 1$ and magnitude greater than $1/4$. Each potential report therefore differs from B with probability less than $\Phi(-3)$. The Gaussian tail estimate

$$\Phi(-x) \leq \frac{e^{-x^2/2}}{x\sqrt{2\pi}}$$

gives $\Phi(-3) < 1/600$: $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$ suffice. Couple the ideal and noisy reports using the same preparation, actual mark and gate random table. Except on a set of probability less than $2\Phi(-3) < 1/300$, both potential reports equal B and the entire gated transcript agrees. The ideal marked law is precisely Q_* , by the prepared (B, W) table. This proves (21) and the feedback-uniform version. Finally, the source is blind, its actual mark is unobserved by the simulator, and the gate is its only current input at the second emission. The product and subfamily assertions follow from the two independent private-emission kernels specified above. $\square$

For $d \leq 2/5$, the two windows instead contain only independent centered Gaussian noise. Their signs are independent fair bits, independent of W , and a private width-one simulator reproduces every gated row exactly. Thus the informative regime and its no-collision control use the same detector, preparation convention and actual mark.

7. THE PHYSICAL EXPERIMENT AND ITS SHARP PREPARATION BOUNDARY

Take $\Omega = \{0, 1\}^3$, with coordinates (x_1, x_2, w) . The all-on private candidate is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad 0 \leq p, q \leq 1.$$

The mark w is the actual mark, not an invented simulator bit. Put $a = (1 - p)(1 - q)$ and $c = pq$. On the ordered atoms

$$(22) \quad 000, \quad 001, \quad 110, \quad 111,$$

let Q_* have masses $5/16, 3/16, 3/16, 5/16$, and set it to zero elsewhere. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. A target Q is allowed whenever $\text{TV}(Q, Q_*) \leq \beta$.

The original feedback operation has the form

$$(23) \quad T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w), \quad g : \{0, 1\} \longrightarrow \{0, 1\}.$$

The selected gate is observable. It may depend on earlier preparations and on the first current report. The target rows are pushforwards by (23). For the converse, restrict the adversary to the admissible private subfamily whose second emission is AX_2 , where A is the current gate and $X_2 \sim \text{Ber}(q)$ is independent of the first emission and of the fair mark. Every row is then a causal garbling of $b(p, q)$. For the construction only the all-on row is used, and every private candidate in the original width-one class has the all-on product law just specified. These two facts, proved for the microscopic interface in Definition 6.1 and Theorem 6.3, explain why using a subfamily for the lower bound does not narrow the upper construction.

The admissible audit first trains, then selects one row–event pair, then uses one fresh candidate and one fresh target validation. It reports $D = \mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The event and row cannot be reselected after seeing a validation draw. All bit, mark, decision and validation cuts are charged. Define $N_c^{\text{phys}}(W, Q)$ to be the minimum number of candidate training preparations among these original-interface auditors, with peak register width at most W , whose score is uniformly strictly greater than c . Randomization, feedback and stopping are allowed. If no such audit exists, the value is $+\infty$.

Theorem 7.1 (Two-preparation physical segment). *Let u_1, m_2 be as in (1). For every $\beta \geq 0$, $\text{TV}(Q, Q_*) \leq \beta$, and*

$$W \geq 12, \quad u_1 + \beta \leq c < m_2 - \beta,$$

one has $N_c^{\text{phys}}(W, Q) = 2$. The two-preparation audit is deterministic, uses no acquired calibration and no random bits, and has the full profile

$$(24) \quad (1, 2, 3, 3, 4, 5, 5, 10, 12, 10, 10, 7, 3)$$

from the initial cut through its twelve bit/mark updates. In particular, its decision alphabet has five labels and its peak has twelve. The lower bound remains valid with no memory cap.

The proof occupies the next three subsections. The interval in the statement can be empty for large β ; no assertion is then made. For $\beta = 1/300$, it contains $c = 2/5$ strictly between its endpoints.

7.1. An exact rational certificate against one preparation.

Lemma 7.2. *Every original-interface auditor using at most one candidate training preparation has worst-case score at most $u_1 + \beta$, regardless of memory.*

Proof. By Theorem 3.2, it suffices to bound $U_1(b, Q_*)$ and then perturb the target. In (5) use the four fixed parameters

$$(p, q) = (1/8, 1/8), (1/5, 1/5), (4/5, 4/5), (7/8, 7/8)$$

with weights $1/16, 7/16, 7/16, 1/16$, respectively. This measure is not asserted to be least favorable. Since $p = q$ on its support, the training count $K = x_1 + x_2$ is sufficient. The independent training mark and the order of unequal bits contain no further information about this prior.

For $k = 0, 1, 2$ and the four validation atoms in (22), the signed coefficients in (5), after grouping by K , are the entries of

$$(25) \quad \frac{1}{40960000} \begin{pmatrix} -1977 & -1775177 & 2170423 & 3943623 \\ 1765554 & 191954 & 191954 & 1765554 \\ 3943623 & 2170423 & -1775177 & -1977 \end{pmatrix}.$$

For example, the coefficient in column j is

$$\sum_{i=1}^4 \mu_i \binom{2}{k} p_i^k (1-p_i)^{2-k} \left(Q_{*,j} - \frac{(1-p_i)^2 \mathbf{1}_{j \leq 2} + p_i^2 \mathbf{1}_{j \geq 3}}{2} \right).$$

This identity proves every entry of (25) by rational arithmetic. All coefficients at off-diagonal validation atoms are nonpositive. The sum of the positive entries in (25) is $4035777/10240000 = u_1$. Each response coordinate is in $[0, 1]$, so this sum bounds its prior-averaged score and hence its minimum over the four fixed candidates.

Replacing Q_* by Q changes the score of any trained response by at most β . The training law is a candidate law and is unchanged. Consequently the same prior bounds every score by $u_1 + \beta$. A procedure with no training, or one that stops before taking its permitted sample, is covered by ignoring the generated all-on observation. $\square$

7.2. A marked five-event selector. Take two independent all-on training preparations. Let S be the total number of ones among their four report bits; discard their marks. Choose the following events, represented by their indicators on (22) and with indicator zero at every off-diagonal atom:

$$(26) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & 0 \end{array}$$

Denote these indicators by $h_0, \dots, h_4$ and report $D = h_S(Y) - h_S(X)$ on independent validation draws.

Lemma 7.3 (Exact uniform score of the selector). *Against the ideal target, the selector (26) has uniform minimum score $m_2 = (12 - 3\sqrt{3})/16$. Against Q , its score is at least $m_2 - \beta$.*

Proof. The generating polynomial of S is

$$((1-p) + pz)^2((1-q) + qz)^2.$$

Writing $g_s = (Q_* - b(p, q))h_s$, its expected score is $G(p, q) = \sum_{s=0}^4 [z^s]((1-p) + pz)^2((1-q) + qz)^2 g_s$. Substitution of the five rows in (26) and collection of terms with $x = p + q - 1$, $y = p - q$, $u = x^2$ and $v = y^2$ gives

$$(27) \quad \begin{aligned} G(p, q) = H(u, v) = & -\frac{u^3}{32} + \frac{3u^2v}{32} + \frac{3u^2}{16} - \frac{3uv^2}{32} - \frac{uv}{4} - \frac{3u}{32} \\ & + \frac{v^3}{32} + \frac{v^2}{16} + \frac{15v}{32} + \frac{7}{16}. \end{aligned}$$

This is a polynomial identity, not an interpolation over a grid. Since $0 \leq u \leq 1$ and $v \geq 0$ on the parameter square,

$$\partial_v H(u, v) = \frac{3(u-v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}.$$

Thus $H(u, v) \geq H(u, 0) = (-u^3 + 6u^2 - 3u + 14)/32$. The derivative of the last expression is $(-3u^2 + 12u - 3)/32$. On $[0, 1]$ it changes sign from negative to positive at $u = 2 - \sqrt{3}$. Substitution gives $(12 - 3\sqrt{3})/16$. The minimizing values are feasible: take $p = q = (1 \pm \sqrt{2 - \sqrt{3}})/2$. This proves exactness of the minimum for this specified selector, not optimality of its score over all

selectors with two preparations. Finally, for each s , the target expectation of $h_s \in [0, 1]^\Omega$ changes by at most β . Averaging over training proves the claimed robustness. $\square$

7.3. Causal compilation and the full register profile.

Lemma 7.4. *The selector and its two validations admit the deterministic profile (24). No selector, first-bit buffer or validation accumulator is outside the charged register. An autonomous phase-tagged realization uses at most 75 states.*

Proof. After each training report bit retain the count accumulated so far. The two marks are read and discarded. The first six updates therefore have widths 2, 3, 3, 4, 5, 5 after the initial singleton. At the decision cut the five count values select the five events.

During candidate validation, after the first bit retain the event and that bit, using at most ten states. After the second bit retain the event and the function of the unread mark that gives its candidate-event indicator. For events 0, 1, 2, 3, 4, respectively, the possible functions are

$$\{0, 1\}, \quad \{0, 1 - w, 1\}, \quad \{0, 1\}, \quad \{0, w, 1\}, \quad \{0, 1\}.$$

There are twelve event–function pairs. After the candidate mark, retain the event and its now known candidate indicator $c_0 \in \{0, 1\}$, using at most ten states.

During target validation, after the first bit discard the event name and retain c_0 and its residual function of the second bit and mark. These residual functions belong to

$$\{0, x_2, (1 - x_2)(1 - w), 1 - x_2, x_2 w\}.$$

This uses at most ten states. After the second bit, the residual *difference* as a function of the unread mark belongs to

$$\{-1, 0, 1, w, 1 - w, w - 1, -w\},$$

so seven states suffice. The mark update gives the three possible outputs $-1, 0, 1$. Each transition depends only on its retained residual and the current report. Thus it is a legal common observable row, not a family of independently optimized hidden-history maps.

For completeness, equality of two residual functions implies equality of their restrictions after either next bit, so identifying equal residuals is closed under every subsequent update. The supplied transition tables are obtained by these restrictions and can be reconstructed directly from (26). Enumerating residuals is a convenience for reproducing the tables, not the proof of their sufficiency. Finally, disjointly tag the states at the thirteen cuts by their phase and use terminal self-loops. The sum of the displayed widths is 75. This implementation charges the clock; it does not assert that 75 is optimal. $\square$

Proof of Theorem 7.1. Lemma 7.2 excludes $N \leq 1$ for a strict score greater than $c \geq u_1 + \beta$, for every architecture and every memory budget. Lemmas 7.3 and 7.4 supply a two-preparation architecture of peak twelve with score at least $m_2 - \beta > c$. Larger width budgets can use the same machine. $\square$

Corollary 7.5 (The positive-noise collision family). *For the original marked collision targets with $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$, the local microscopic estimate $\text{TV}(Q_{d,\sigma}, Q_*) < 1/300$ implies*

$$N_{2/5}^{\text{phys}}(W, Q_{d,\sigma}) = 2 \quad (W \geq 12).$$

There are four preparations in the complete audit: two candidate training, one candidate validation, and one target validation.

Proof. Apply Theorem 7.1. The lower-end margin is the exact rational number

$$\frac{2}{5} - u_1 - \frac{1}{300} = \frac{78269}{30720000} > 0.$$

For the other endpoint, $\sqrt{3} < 7/4$ implies $m_2 > 27/64$ and $27/64 - 1/300 > 2/5$. The physical total-variation estimate, including the actual mark and positive sensor noise, is proved in Theorem 6.3; the current deduction does not replace those hypotheses by a zero-noise limit. $\square$

Remark 7.6 (The part of the frontier determined here). The theorem evaluates the minimum preparation coordinate for $W \geq 12$. It does not determine $N_c^{\text{phys}}(W, Q)$ for $3 \leq W \leq 11$, the maximum score with two preparations, or the least peak width at $N = 2$. The earlier exact decision-width result $\kappa_{2/5}^{\text{rows}} = 3$ is unchanged: it allows longer training, whereas this shortest audit uses five decision states. The older points (399, 40602), (320000, 1604) and $(12800 \cdot 2^{400}, 12)$ remain valid constructions with decision width three. They are dominated by (2, 12) in the projection onto training preparations and peak width, but not necessarily in a resource order that also constrains the decision alphabet separately.

Corollary 7.7 (The enlarged physical preparation segment). *With the full architecture of Definition 6.1,*

$$N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad \text{TV}(Q, Q_*) \leq \beta, \quad u_{\text{ex}} + \beta \leq c < m_2 - \beta.$$

Here $u_{\text{ex}} = (85 - 7\sqrt{73})/64$ is the exact one-preparation all-on value, rather than the older rational upper bound.

Proof. Restrict the adversary to the common-garbling subfamily in Definition 6.1. Theorem 3.2 and Corollary 5.2 bound every at-most-one-preparation auditor, even with unlimited memory, by $u_{\text{ex}} + \beta$. The same deterministic two-preparation machine in Lemmas 7.3 and 7.4 has score at least $m_2 - \beta$ against the full private candidate family. These inequalities prove the assertion. The four-preparation total, peak twelve and positive-noise specialization are unchanged. $\square$

8. AN EXACT PREPARATION–DECISION-MEMORY LAW

A second solvable family separates a genuine joint resource law from an implementation bound. Here the memory coordinate is the number of states at the training-validation cut. It is not identified with the bit-level peak width of the collision audit.

Fix $m = 2^d$, $0 < \alpha < 1$, and an unknown parameter $\theta \in \{1, \dots, m\}$. A fresh training preparation reports θ with probability α and an erasure $\perp$ otherwise. Conditional on θ , the preparations are independent. Candidate validation has law δ_θ and target validation is uniform on $\{1, \dots, m\}$. At the decision cut the auditor retains at most $K = 2^k$ states, $0 \leq k \leq d$. Its subsequent response may depend only on that state, current validation observations and fresh independent randomness. In particular, a calibration table, remembered seed or stopping time outside the retained state is not available. Upstream and downstream memory are otherwise unrestricted for the converse. Let $V_N^{\text{er}}(K)$ be the best worst-case expected signed validation score with at most N training preparations.

Theorem 8.1 (Exact joint law). *For every $N \geq 0$ and $K = 2^k \leq m$,*

$$(28) \quad V_N^{\text{er}}(K) = (1 - (1 - \alpha)^N) \left(1 - \frac{1}{K}\right).$$

The converse covers stochastic encoders and arbitrary legal stopping. The value is attained by a clocked machine with K training states. With atomic preparation and validation outcomes, the complete audit has peak at most $\max\{2K, 3\}$ and uses at most $N + 2$ preparations.

Proof. Write $q = (1 - \alpha)^N$ and pre-generate N potential training reports. A stopped auditor ignores the unused reports; this does not disclose extra information to it. On the event that all N reports are erasures, its retained-state law $r = (r_j)_{j=1}^K$ is independent of θ . After padding any smaller register with unused states, its channel to the decision register has the form

$$(29) \quad C(j \mid \theta) = qr_j + (1 - q)S(j \mid \theta)$$

for a stochastic channel S . This identity remains valid with randomized stopping and randomized state updates.

Use the uniform prior on parameters as an upper certificate. The part qr_j has zero expected score for every response: averaging δ_θ under this prior gives the target itself. For the other part, optimizing over all responses gives a convex function of S , a sum of positive parts of linear functions. Every stochastic channel is a convex combination of deterministic channels $j = g(\theta)$; therefore its value is at most the largest deterministic-channel value. This convexification is used only to bound a channel. It does not presuppose that a mixture of width- K machines is realizable at width K .

If the fibers of g have sizes $n_1, \dots, n_K$, a nonempty fiber has posterior uniform on that fiber, so its total variation from the uniform target is $1 - n_j/m$. The prior-averaged optimal score is

$$\sum_j \frac{n_j}{m} \left(1 - \frac{n_j}{m}\right) = 1 - \sum_j (n_j/m)^2 \leq 1 - 1/K.$$

Empty fibers contribute zero. Cauchy–Schwarz proves the inequality. This is an upper bound on the average score and hence on its minimum over fixed parameters. Multiplying by $1 - q$ proves the upper bound in (28).

For attainment, partition the parameter alphabet into K equal groups and let $g(\theta)$ denote its group. Initialize the register uniformly on the K group labels. On an erasure leave it unchanged; on a revealed parameter overwrite it by $g(\theta)$. No extra “revelation occurred” flag is needed. At the decision cut its law is $q \text{Unif}[K] + (1 - q)\delta_{g(\theta)}$. For register j , freeze the event which is the complement of group j . On a revelation its score is $1 - 1/K$. Under a uniformly random initial label, the mean score is zero for each fixed θ : one of the K labels has score $1 - 1/K$ and the others have score $-1/K$. This proves the exact worst-case value.

The k initialization bits can be accumulated in the K -state register; all earlier prefix counts are at most K . Thus no persistent seed lies outside it. Candidate validation retains the group label and its event indicator, using at most $2K$ states. Target validation produces one of the three values $-1, 0, 1$. This proves the atomic-outcome peak bound. The controller uses a declared external schedule; an autonomous stopping implementation must additionally retain its phase. A phase-tagged implementation is finite, but its cost is not included in K . $\square$

Corollary 8.2 (Thresholds and matching resource scales). *For $K \geq 2$ and $0 \leq c < 1 - 1/K$, the least integer number of training preparations giving a score strictly greater than c is*

$$(30) \quad N_c^{\text{er}}(K) = \left\lceil \frac{\log(1 - c/(1 - 1/K))}{\log(1 - \alpha)} \right\rceil + 1.$$

If $c \geq 1 - 1/K$, no finite N suffices. For $0 < \varepsilon < 1$, achieving $V_N^{\text{er}}(K) \geq 1 - \varepsilon$ requires

$$K \geq \varepsilon^{-1}, \quad N \geq \frac{\log(1/\varepsilon)}{-\log(1 - \alpha)}.$$

Conversely, whenever an allowed $K \leq m$ satisfies $K \geq 2/\varepsilon$, the choice $N \geq \log(2/\varepsilon)/[-\log(1 - \alpha)]$ suffices.

Proof. The strict threshold is equivalent to $(1 - \alpha)^N < 1 - c/(1 - 1/K)$. Since the denominator logarithm is negative, the least integer satisfying this strict inequality is (30), including equality cases of the real quotient. Also

$$1 - V_N^{\text{er}}(K) = q + (1 - q)/K.$$

This is at least q and at least $1/K$, and is at most $q + 1/K$. These three inequalities prove the necessary and sufficient bounds. $\square$

The exact law concerns an entire (N, K, α, m) family, not a single finite diagnostic. Its mechanism is a parameter-independent erasure component followed by the convex geometry of the decision channel. It demonstrates how a preparation obstruction and a memory obstruction can be matched in one theorem. It does not identify the least physical peak width at two collision preparations. Encoding the m -symbol reports bit by bit also introduces cuts not present in this atomic model; the physical bit-level accounting is given separately.

9. QUANTITATIVE CERTIFICATES FOR A FINITE COMPATIBLE ARCHITECTURE

We return to an arbitrary finite architecture, now with a finite parameter set. It can have hidden states, controlled experiments, more than two hypotheses and registers of any size. Enumerate its distinct stochastic rows by $r = 1, \dots, R$ and write $q_r \in \Delta_{s_r}$. A reused autonomous row appears only once in this product. Assume each execution uses at most H controller rows, and uses row r at most d_r times. Delete unused rows and put $S = \max_r s_r$. Fixed environmental kernels are not controller rows.

For an integer $m \geq 1$, intersect Δ_{s_r} with the coordinate boxes of side $1/m$. The resulting nonempty closed polytopes cover the simplex, with overlaps permitted at their boundaries. Write C for a product of such row cells and choose a vertex q_r^C in each factor. A row-cell vertex has at most one coordinate not fixed at a box boundary; consequently it has at most $v_r = s_r 2^{s_r - 1}$ vertices, all on the $1/m$ grid. There are at most $m^{\sum_r s_r}$ product cells.

Write the vertices of row cell C_r as $v_{r,j}$ and parametrize $q_r = \sum_j \lambda_{r,j} v_{r,j}$ with λ_r in a simplex. A reward polynomial has degree at most d_r in block λ_r . Use the homogeneous simplex Bernstein basis of degree d_r in that block, obtaining

$$(31) \quad g_\theta(q(\lambda)) = \sum_\alpha b_{\theta,C,\alpha} B_\alpha(\lambda), \quad B_\alpha \geq 0, \quad \sum_\alpha B_\alpha = 1.$$

Degree elevation includes the factors $\sum_j \lambda_{r,j} = 1$. No independence between different visits of a shared row is imposed on its unknown row vector.

Lemma 9.1 (Causal interpretation of the coefficients). *Every coefficient in (31) satisfies*

$$|b_{\theta,C,\alpha} - g_\theta(q^C)| \leq HS/m.$$

Proof. For each row r , independently prepare an ordered list of d_r vertex labels, sampled independently with law λ_r . At its k th visit, use the stochastic row given by the k th vertex label in that list. Before conditioning, the next label is independent of the previous execution, so this experiment has exactly the row q_r at every visit and expected reward $g_\theta(q(\lambda))$. Unused labels may be sampled after termination without altering the reward.

Condition on the counts of every vertex label in every list. The probability of those counts is $B_\alpha(\lambda)$. Conditional on the counts, each list is a uniformly ordered multiset. Its expected reward is independent of λ and is therefore the corresponding Bernstein coefficient. This also proves (31) directly.

Each row used by this conditioned experiment is a vertex of C_r or a mixture of its remaining vertices. Every two vectors in C_r have coordinate differences at most $1/m$, hence total variation

at most $s_r/(2m) \leq S/(2m)$. Couple the conditioned experiment to the one using q^C , using the same environmental randomness while their executions agree. At each actual controller update the probability of a first row-level disagreement is at most $S/(2m)$. There are at most H such updates on the agreeing path. The terminal laws consequently differ in total variation by at most $HS/(2m)$. A reward in $[-1, 1]$ has expectation difference at most twice that quantity. This proves the estimate. The conditioned urn is used to analyze a coefficient; it is not declared to be an admissible free-memory controller. $\square$

Theorem 9.2 (Explicit subdivision certificate). *Define*

$$\ell_C = \min_{\theta} g_{\theta}(q^C), \quad u_C = \min_{\theta} \max_{\alpha} b_{\theta, C, \alpha}, \quad L_m = \max_C \ell_C, \quad U_m = \max_C u_C.$$

For the exact all-stochastic value of the specified architecture,

$$(32) \quad L_m \leq V(A) \leq U_m \leq L_m + HS/m.$$

On each cell, a witnessing parameter θ_C supplies the nonnegative coefficient certificate

$$u_C - g_{\theta_C}(q(\lambda)) = \sum_{\alpha} (u_C - b_{\theta_C, C, \alpha}) B_{\alpha}(\lambda).$$

Its degree in row block r is d_r , and its total degree is at most $\sum_r d_r$. At most $\prod_r \binom{d_r + v_r - 1}{v_r - 1}$ coefficients per parameter and cell are needed. If the primitive data are rational, all coefficients and bounds can be checked by rational arithmetic. Taking $m \geq HS/\varepsilon$ gives a certified interval of length at most ε .

Proof. The point q^C is feasible, so ℓ_C is a lower bound on the value restricted to C . For any $q \in C$, choose a convex representation of its rows. Equation (31) implies

$$\min_{\theta} g_{\theta}(q) \leq \min_{\theta} \max_{\alpha} b_{\theta, C, \alpha} = u_C.$$

Choose a parameter attaining the minimum defining ℓ_C and apply Lemma 9.1; this gives $u_C \leq \ell_C + HS/m$. Taking the maximum over the finite cover proves (32). Choose θ_C attaining the definition of u_C ; all displayed coefficients of its certificate are nonnegative. The degree and count statements follow from the product Bernstein basis. Cell vertices are rational grid points, and a finite execution sums finite products, proving the rational statement. $\square$

Remark 9.3. The number of cells and coefficients can be enormous. Neither polynomial running time nor a uniform degree bound for the original sum-of-squares hierarchy follows. Refinement changes the domain cover, not merely a single global SOS degree. For a finite collection of architectures obeying a specified budget, maxima of these intervals give outer intervals; an unbounded architecture class needs a separate uniform argument such as Theorem 3.2. For compact parameter families, an effective net and a verified modulus for the primitive kernels are additional inputs, not consequences of compactness alone.

The resource estimate also makes row precision explicit. With $m = 2^b$, the lower witnesses have dyadic rows and lose at most $HS2^{-b}$ in value. Thus increasing row precision provides a quantified approximation for a fixed architecture, while its random-bit implementation still has to be charged. The physical theorem avoids this cost altogether because its rows are deterministic.

10. AUTONOMOUS NOISY TESTING WITH A PRECISION AND TIME LEDGER

This section supplies a substantive autonomous comparison rather than merely identifying shared rows. The stationary optimum is classical [6]; the proof and explicit finite-time implementation are recorded to make the resource conventions unambiguous. They are not used as novelty evidence for the physical theorem.

Let reports be independent, with law $\text{Ber}(p)$ under hypothesis $-$ and $\text{Ber}(q)$ under hypothesis $+$, where $0 < p < 1/2$, $q = 1 - p$, and $r = q/p$. A W -state autonomous machine has two time-independent stochastic transition matrices T_0, T_1 . Assume their induced chains $P_- = qT_0 + pT_1$ and $P_+ = pT_0 + qT_1$ are irreducible. A final decision uses only the state, with optional fresh terminal randomization.

Lemma 10.1 (Likelihood-range bound). *If equivalent probability laws ν_+, ν_- on a finite set satisfy*

$$\frac{\max_i \nu_+(i)/\nu_-(i)}{\min_i \nu_+(i)/\nu_-(i)} \leq R,$$

then $\text{TV}(\nu_+, \nu_-) \leq (\sqrt{R} - 1)/(\sqrt{R} + 1)$.

Proof. Write $L = d\nu_+/d\nu_-$, $a = \min L$ and $b = \max L$. Since $\mathbb{E}_{\nu_-} L = 1$, one has $a \leq 1 \leq b$. The chord bound for $(L - 1)_+$ gives

$$\text{TV}(\nu_+, \nu_-) = \mathbb{E}_{\nu_-} (L - 1)_+ \leq \frac{(1 - a)(b - 1)}{b - a}.$$

For fixed b/a , maximizing the last expression gives $a = 1/\sqrt{b/a}$; substitution yields $(\sqrt{b/a} - 1)/(\sqrt{b/a} + 1)$. It is increasing in b/a , proving the assertion. The identical-law case follows directly. $\square$

Theorem 10.2 (Autonomous stationary bound and an explicit approximation). *Put $D = W - 1$, $W \geq 2$. In the irreducible autonomous class the infimum stationary minimax error is*

$$(33) \quad e_W = \frac{1}{1 + r^D}.$$

For every $\epsilon = 2^{-b}$, $b \geq 0$, there is a machine whose rows have denominator dividing 2^{b+1} , and whose minimax error at report time n from any initial state is at most

$$(34) \quad e_W + \epsilon(W - 2) + \exp \left\{ - \left(\frac{p\epsilon}{2} \right)^D \left\lfloor \frac{n}{D} \right\rfloor \right\}.$$

The stationary lower bound is not a finite-horizon lower bound.

Proof. Every positive induced transition has its likelihood ratio in $[r^{-1}, r]$, and the supports of the two induced matrices are identical. The Markov chain tree formula writes each stationary weight as a sum of products over directed spanning trees rooted at that state. Each product has D transitions; its ratio lies in $[r^{-D}, r^D]$. The ratio of the two unnormalized stationary weights lies in the same interval. Normalizing multiplies all state likelihood ratios by a common number, so their maximum-to-minimum ratio is at most r^{2D} . Lemma 10.1 therefore bounds stationary total variation by $(r^D - 1)/(r^D + 1)$. The equal-prior Bayes error is at least $1/(1 + r^D)$, and a minimax error cannot be smaller than that Bayes error. This is the equal-prior binary specialization of the Hellman–Cover bound.

For the upper approximation, take states $i = 0, \dots, D$, weights $c_0 = c_D = 1$, and $c_i = \epsilon$ at interior states. On report 1, move from i to $i + 1$ with probability $a_i = \frac{1}{2} \min(1, c_{i+1}/c_i)$, if that neighbor exists; otherwise stay. On report 0, move to $i - 1$ with probability $b_i = \frac{1}{2} \min(1, c_{i-1}/c_i)$, if it exists; otherwise stay. All other probability is assigned to staying. Detailed balance gives

$$\pi_+(i) = \frac{c_i r^i}{\sum_j c_j r^j}, \quad \pi_-(i) = \frac{c_i r^{-i}}{\sum_j c_j r^{-j}}.$$

Reflection $i \mapsto D - i$ exchanges these laws. Decide $+$ for $i > D/2$, $-$ for $i < D/2$, and use a fair terminal decision at $i = D/2$ when this state exists. The two errors agree. Conditional on being

at an endpoint, the error is exactly e_W . The total interior stationary mass is at most $\epsilon(W - 2)$, since each $r^i \leq r^D$ and the denominator is at least r^D under $+$, with the reflected argument under $-$. This proves the first two terms of (34).

Every existing neighbor transition under either hypothesis has probability at least $p\epsilon/2$, and every self-loop has probability at least $1/2$. From any state, move down to zero and then stay until D steps have elapsed. This yields the common minorization $P_{\pm}^D(i, \{0\}) \geq \delta = (p\epsilon/2)^D$. Subtracting that common atom gives contraction in total variation by $1 - \delta$ every D steps. Comparison with the stationary law yields $\text{TV}(P_{\pm}^n(i, \cdot), \pi_{\pm}) \leq (1 - \delta)^{\lfloor n/D \rfloor} \leq e^{-\delta \lfloor n/D \rfloor}$. A bounded decision changes error by at most this amount, proving (34). Finally, letting n tend to infinity and then ϵ to zero proves that the stationary infimum equals (33). All transition probabilities belong to $\{0, 1, 1/2, \epsilon/2, 1 - 1/2, 1 - \epsilon/2\}$. $\square$

The W states in this theorem refer to an atomic stochastic update after each report. A sequential fair-bit implementation must also retain the report, the current state, a bit counter and a flag. Each nontrivial move probability is $1/2$ or $2^{-(b+1)}$; test for a string of zero bits of the required length, and pad to $b + 1$ bits if a uniform report cycle is required. At most $4W(b + 2)$ states suffice for this implementation, including its bit-phase counter and a ready phase. This conservative bound is an implementation claim, not an optimal precision–state tradeoff. The time index n in (34) is an observation time at which an external analyst reads the autonomous machine, not a free input clock used by its transitions. The adversarial model of [10] is not covered by the independent-report proof above.

11. PHYSICAL CONSUMERS AND THE FIRST-PRINCIPLES CONNECTION

The physical preparation theorem needs no estimate of the target law as a stored input: the five events are fixed and their uniform validity uses a proved neighborhood of Q_* . This removes calibration memory from this particular consumer without asserting that target acquisition is unnecessary in an arbitrary comparison problem. The general acquired calibration theorem, with its full counter register, remains in the complete companion.

Corollary 11.1 (A raw-interface confidence construction). *Let $m = m_2 - \beta > 0$ and $0 < c < m$. Repeat the full four-preparation audit independently R times and write $\overline{D}_R$ for the mean score. Under every private candidate,*

$$\mathbb{P}(\overline{D}_R \leq c) \leq \exp\{-R(m - c)^2/2\}.$$

Under the null in which the independent candidate and target validation laws agree, with the same event chosen before both validations,

$$\mathbb{P}(\overline{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

A clocked implementation uses $4R$ preparations and at most $12(2R + 1)$ persistent states, with no calibration and no random bits.

Proof. The independent block scores lie in $[-1, 1]$ and have mean at least m under a private candidate. Hoeffding's inequality gives the first bound. Under the null, conditional on the selected event, the two independent validation indicators have the same mean. Each block therefore has mean zero, and the same inequality gives the second bound. Retain the integer sum in $\{-R, \dots, R\}$ along with the current twelve-state block. At a block boundary add its terminal score and reset its register. The deterministic external schedule is part of the clocked convention. $\square$

This is an upper construction on the raw marked interface. Its proof does not transfer a lower bound from a deliberately reduced Bernoulli interface, and no optimal confidence-memory frontier is claimed.

Proposition 11.2 (Target transport without a training multiplier). *For the two-preparation audit, changing only the target from Q to Q' changes its expected score by at most $\text{TV}(Q, Q')$. Its candidate training law, register profile and event selector remain unchanged.*

Proof. Condition on the candidate training word. The expectation difference is $(Q - Q')h_S$, whose absolute value is at most $\text{TV}(Q, Q')$. Average over that unchanged training law. $\square$

In particular, every common-preparation transport estimate verified for the original positive-noise collision family can be inserted here without multiplying it by a long training horizon. If candidate kernels also change, their training and validation defects must be added by a causal coupling; the target-only statement does not remove those terms.

The first-principles route is now concrete. The positive experimental kernels and actual reset preparations give the likelihoods in (5); these generate posterior continuation states. The validation-test quotient gives the positive parts in the statistical dual. A prior then supplies an architecture-uniform impossibility certificate. On the constructive side, the five-event decision is compiled by taking its future residual functions, which are closed under observation shifts. These are compatible causal states and determine the charged profile. Thus the preparation law, predictive continuation and finite-resource realization are used in the same upper/lower theorem. This is not a deduction of geometry or nonlinear operator compactness from the existence of finite stochastic rows.

12. DEPENDENCIES, RETAINED RESULTS AND REMAINING DISTINCTIONS

The exact three-report binary noisy frontier, arbitrary-horizon binary-register minimax recursion and compatible Bayesian interval recursion remain as proved in the complete companion. Its exact odd-horizon majority realization is also retained; it is not a new result of this revision. Section 9 applies to registers of any finite width, controlled rows and any finite parameter set, but provides certified intervals rather than a closed minimax recursion for all such classes. The physical theorem supplies a matched preparation segment in a continuous two-parameter family with overlapping stochastic reports; its converse is not a finite parameter-grid check.

The stationary autonomous result in Section 10 has an explicit formula, but its stationary lower bound must not be applied to a finite-horizon clocked machine. Its finite-time upper bound is purposely conservative and exposes the slowing caused by finite transition precision. An arbitrary compact-family presentation still requires additional effective information to construct a kernel net. A subdivided positive-basis degree bound must not be called a degree bound for a global SOS hierarchy.

The physical result is a genuine consumer of the experimental and continuation root. It does not make the independent A2 geometric proofs dependent on this root. The original B4 normalized-resolvent, range, compactness and control-transfer obligations, and the broad C2 strict-dual, form, rigidity and optional-projection obligations, are not proved by our finite-state theorem. Their statements, derivations and current status remain in the retained historical pipeline. No aggregate closure is inferred from preserving their source files.

Finally, the rational certificate (25), polynomial identity (27), and residual transitions are independently reproducible. The program in the accompanying source checks their exact identities and includes deliberately invalid controls. These checks are not substitutes for the proofs above. Neither compilation, preservation of earlier pages nor a successful finite test establishes novelty or an editorial decision.

REFERENCES

- [1] R. D. Smallwood and E. J. Sondik, The optimal control of partially observable Markov processes over a finite horizon, *Operations Research* **21** (1973), 1071–1088.
- [2] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272.
- [3] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176.
- [4] J. Müller and G. Montúfar, The geometry of memoryless stochastic policy optimization in infinite-horizon POMDPs, arXiv:2110.07409 (2021; revised versions).
- [5] M. Putinar, Positive polynomials on compact semi-algebraic sets, *Indiana Univ. Math. J.* **42** (1993), 969–984.
- [6] M. E. Hellman and T. M. Cover, Learning with finite memory, *Ann. Math. Statist.* **41** (1970), 765–782.
- [7] M. E. Hellman and T. M. Cover, On memory saved by randomization, *Ann. Math. Statist.* **42** (1971), 1075–1078.
- [8] R. A. Flower and M. E. Hellman, Hypothesis testing with finite memory in finite time, *IEEE Trans. Inform. Theory* **18** (1972), 429–431.
- [9] T. M. Cover, M. A. Freedman and M. E. Hellman, Optimal finite memory learning algorithms for the finite sample problem, *Information and Control* **30** (1976), 49–85.
- [10] M. A. Managoli and V. M. Prabhakaran, Memory constrained adversarial hypothesis testing, arXiv:2605.12063v1 (May 12, 2026).
- [11] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28, doi:10.1524/strm.2002.20.14.1.
- [12] Q. Qi, General Theta Foundations I: Compatible statistical experiments and resource frontiers, revision 23, September 24, 2026, complete mathematical companion, source commit 8a6043aca644043bd805668185eb227468f21170; packaged commit e926b783b643b3e7f275a18105528927b76cc97b. The 96-page article and 649-page development are reproduced without alteration after the present article in the two complete volumes.